

KARAN RAI

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Professional Summary

Aspiring **Software Engineer** with hands-on experience in **Full-Stack Development** using React.js, Node.js, and Express.js. Strong foundation in **Data Structures, Algorithms, Object-Oriented Programming, and DBMS**. Experienced in building scalable web applications, RESTful APIs, and real-time systems. Passionate about solving real-world problems with efficient and maintainable solutions.

Technical Skills

Programming Languages: Python, Java, C++, JavaScript (ES6+), SQL
Web Technologies: React.js, Node.js, Express.js, REST APIs, HTML5, CSS3
Databases: MySQL, PostgreSQL
Core Concepts: Data Structures, Algorithms, OOP, DBMS, Computer Networks
Tools & Platforms: Git, GitHub, Firebase

Projects

Automotive Service Management System

Oct 2024

- Developed a **full-stack web application** using React.js, Node.js, and Express.js to connect users with nearby automotive service providers.
- Designed and implemented **RESTful APIs** for authentication, booking workflows, and service management.
- Built an intelligent **appointment scheduling system**, reducing manual booking time by **40%**.
- Integrated **Twilio API** to enable real-time SMS notifications and service reminders.
- Developed a **rule-based pricing engine** to automate cost calculation, reducing billing disputes by **75%**.
- Ensured **secure data handling** and optimized backend performance using middleware and validation.

Automated Photo Organization using Machine Learning

Feb 2025

- Built an end-to-end **machine learning pipeline** using CNN models for image classification and tagging.
- Implemented **clustering algorithms** for duplicate image detection with **92% accuracy**.
- Designed a **quality assessment module** to filter low-quality images with **88% precision**.
- Applied **feature extraction techniques** to improve model efficiency and reduce processing time.
- Developed a **responsive React.js interface** for seamless user interaction.
- Improved workflow efficiency by **65%** through automation and intelligent sorting.

InteliQR – Emergency Medical Response System

Jun 2025

- Designed an **IoT-enabled emergency response system** integrating QR codes with real-time GPS tracking.
- Built a **real-time SOS alert system** using Node.js and Firebase for instant communication.
- Implemented **secure authentication and authorization** to protect sensitive user data.
- Enabled **automatic location sharing**, reducing emergency response time by **60%**.
- Designed scalable backend architecture ensuring **high availability and reliability**.
- Focused on **low-latency API handling** to improve system responsiveness.

Education

VIT Bhopal University, Madhya Pradesh

Sep 2023 – Jun 2027

Bachelor of Technology in Computer Science

Relevant Coursework: Data Structures, Algorithms, DBMS, Computer Networks, Machine Learning

Shri Guru Nanak Higher Secondary School, MP

2023

Senior Secondary (81.5%)

Certifications

Google IT Support Professional Certificate, 2026

Credential Link

Topics Covered: IT Support, Networking, System Administration, Security, Troubleshooting

Data Structures and Performance, 2025

Credential Link

Topics Covered: Arrays, Linked Lists, Trees, Graphs, Time Complexity, Algorithm Optimization

The Bits and Bytes of Computer Networking ,2025

Credential Link

Topics Covered: OSI Model, TCP/IP, DNS, HTTP, Network Protocols, Troubleshooting